

Anthony Hana

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EDUCATION

McMaster University

Sept 2022 - Present (Exp. Apr 2027)

Honours Mathematics and Computer Science (B.Sc.)

Hamilton, ON

- **Extra Curriculars:** McMaster Start Coding, McMaster Game Dev Club, Volleyball, Soccer
- **Courses:** Software Development, Data Structures & Algorithms, Natural Language Processing, Numerical Linear Algebra, Mathematical Modeling, Databases, Computer Architecture

EXPERIENCE

Software Engineer

September 2025 - Present

LTV.ai

Toronto, ON

- Redesigned analytics pipelines and query patterns over **1.1B+ rows of event data**, reducing dashboard load times from **30+ minutes to seconds** and restoring usability at scale
- Designed and launched a self-serve campaign platform, eliminating a manual CS workflow (1+ hour per request) and enabling scalable customer usage contributing to **7-figure ARR**
- Scaled distributed messaging infrastructure by **400%+** through async processing, batching, and worker orchestration, while significantly reducing send failures and bounce rates

Software Architect

June 2025 - Present

STaBL Foundation (McMaster Start Coding)

Hamilton, ON

- Developing the Stabl.Rocks IDE, supporting workshops and university coursework for **80,000+** students
- Shipped custom Haskell plugins and functionality powering MSC's multisite platform and core course workflows
- Reduced latency and improved responsiveness by **10%** across the McMaster Start Coding platform

PROJECTS

LavaLock - DeltaHacks XII | 📱 | 🔒 | Swift, Go, C++, AWS, PostgreSQL

- Won **1st place** at DeltaHacks XII for a hardware password vault powered by **lava lamp** entropy
- Built a native **SwiftUI** iOS app with secure keychain storage and an entropy-driven password generation algorithm
- Architected a **Go/C++** backend processing hardware entropy streams via **AWS S3** and **EC2**

Posture-bot - Hack The North 2025 | 📱 | 🔒 | Python, JavaScript, Typescript, Node.js, DynamoDB

- Built a real-time **posture tracking system** using **OpenCV** and **MediaPipe Pose** to detect postural deviations
- Generated **AI-driven posture reports** summarizing session analytics, joint angle distributions, stability metrics, and personalized improvement insights

TeX-it - HackThe6ix 2025 | 📱 | 🔒 | Electron, React, Typescript, GO, Gemini API

- Engineered a native LaTeX IDE with real-time editing, rendering, and consistent document state across user interactions
- Developed AI pipelines to convert unstructured inputs (voice, image, text) into **compile-ready LaTeX**
- Built validation and feedback mechanisms to ensure **compile-safe LaTeX** and improve editor usability

MacPad - Native MacOS Text Editor | 📱 | Swift, UI/UX, Gemini API

- Built **MacPad**, a native macOS Swift text editor with inline AI-powered writing suggestions
- Optimized editor performance and implemented robust multi-language text encoding for fast, reliable large-file editing

TECHNICAL SKILLS

Languages: Python, Go, C++, Javascript, Typescript, Swift, SQL, Haskell

Frameworks & Tools: Next, Node.js, Flask, REST API's, Git, Github Actions, Docker, AWS, Linux, Angular, Terraform, Jira

ML/Data: PostgreSQL, DynamoDB, Redis, Clickhouse, Kafka, PyTorch, TensorFlow, pandas, scikit-learn, NumPy, OpenCV, Matplotlib

Practices: CI/CD, Unit Testing, Software Development, UI/UX Design, Documentation, OOP, SCRUM/Agile, Debugging, Backend Development, Computer Vision, Cloud Computing, DevOps, LLMops